

Multimedia Application

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- ▶ Conversational Agents
- ▶ Weizenbaum's ELIZA
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Conversational Agents

(AKA Dialogue Systems AKA Dialogue Agents AKA Chatbots)

- ▶ Personal Assistants on phones or other devices
- ▶ SIRI, Alexa, Cortana, Google Assistant
- ▶ Playing music, setting timers and clocks
- ▶ Chatting for fun
- ▶ Booking travel reservations
- ▶ Clinical uses for mental health

Conversational Agents

- ▶ Conversational agents, also called dialogue systems, dialogue agents, chatbots, are **systems designed to interact with humans via conversations**, either **in text or speech**. These include the personal assistants on phones or other devices like SIRI, Alexa, Cortana, or the Google Assistant, tools for simple, relatively short interactions like playing music, setting timers or clocks or keeping shopping lists, to longer conversations maybe just for fun, or for practical applications like booking travel reservations, or even in clinical uses for mental health.

Two kind of conversational agents

1. Chatbots

- mimic informal human chatting
- for fun, or even for therapy

2. (Task-based) Dialogue Agents

- interfaces to personal assistants
- cars, robots, appliances
- booking flights or restaurants

Two kind of conversational agents

Chatbots are systems that can carry on extended conversations with the goal of mimicking the unstructured conversations or 'chats' characteristic of informal human-human interaction. Mostly these have been designed for fun.

- ▶ Yet starting from the very first system, Weizenbaum's ELIZA, the role of chatbots has also been studied for practical purposes like counseling.

Chatbot Architectures

► Rule-based

1. Pattern-action rules ([ELIZA](#))
2. + A mental model ([PARRY](#)):

[The first system to pass the Turing Test!](#)

► Corpus-based

3. Information Retrieval ([Xiaolce](#))
4. Neural encoder-decoder ([BlenderBot](#))

Chatbot Architectures

- ▶ Like practically everything else in language processing, chatbot architectures fall into
- ▶ two classes: **rule-based systems** and **corpus-based systems**. Rule-based systems include the early influential ELIZA and PARRY systems. **Corpus-based systems mine large datasets** of human-human conversations, which can be done by using information retrieval to copy a human response from a previous conversation, or using an encoder-decoder system to generate a response from a user utterance.

BlenderBot (Roller et al. 2020)

Will you sing me a song?

Sure, what do you want to be sung to? I can sing you a song about baking.

Yes, sing me a song about baking!

Do you want me to sing it to the tune of "Sing Along with Mitch"?

Yes, I would love that.

OK, I'll try to get it to you by the end of the day. What else do you like to do?

No, sing it to me now!

OK, give me a few days. I'm a teacher, so I don't have much free time.

Can you at least tell me what the lyrics are?

Sure, it goes like this: "Baking is my favorite thing to do in the world."

How does the chorus go?

It goes: BAKING IS MY FAVORITE THING TO DO IN THE WORLD!

Wow, sounds like a bop.

I know, right? It's so catchy. I can't get it out of my head.

Task-based dialogue agents

- ▶ **Task-based" or "goal-based"** dialogue agents
 - Systems that have the goal of **helping a user solve a task**
 - **Setting a timer**
 - **Making a travel reservation**
 - **Playing a song**
 - **Buying a product**
- ▶ **Architecture:**
 - **Frames with slots and values**
 - **A knowledge structure representing user intentions**

The Frame

- ▶ A set of **slots**, to be filled with information of a given **type**
- ▶ Each associated with a **question** to the user

Slot	Type	Question
------	------	----------

ORIGIN	city	"What city are you leaving from?"
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DEST	city	"Where are you going?"
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DEP DATE	date	"What day would you like to leave?"
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DEP TIME	time	"What time would you like to leave?"
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AIRLINE	line	"What is your preferred airline?"
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Introduction to Chatbots and Dialogue Systems

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₄: And you're flying into what city?
C₅: Seattle.
A₆: And what time would you like to leave Pittsburgh?
C₇: Uh hmm I don't think there's many options for non-stop.
A₈: Right. There's three non-stops today.
C₉: What are they?
A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₁₁: OK I'll take the 5ish flight on the night before on the 11th.
A₁₂: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.
C₁₃: OK.
A₁₄: And you said returning on May 15th?
C₁₅: Uh, yeah, at the end of the day.
A₁₆: OK. There's #two non-stops ... #
C₁₇: #Act...actually #, what day of the week is the 15th?
A₁₈: It's a Friday.
C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK...OK. On Sunday I have ...

A telephone conversation between a human travel agent (A) and a human client (C)

Properties of Human Conversation

► Turns

- We call each contribution a "turn"
- As if conversation was the kind of game where everyone takes turns.

C₁: ...I need to travel in May.

A₂: And, what day in May did you want to travel?

C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.

A₄: And you're flying into what city?

C₅: Seattle.

A₆: And what time would you like to leave Pittsburgh?

C₇: Uh hmm I don't think there's many options for non-stop.

A₈: Right. There's three non-stops today.

C₉: What are they?

A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.

C₁₁: OK I'll take the 5ish flight on the night before on the 11th.

A₁₂: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.

C₁₃: OK.

A₁₄: And you said returning on May 15th?

C₁₅: Uh, yeah, at the end of the day.

A₁₆: OK. There's #two non-stops ... #

C₁₇: #Act...actually #, what day of the week is the 15th?

A₁₈: It's a Friday.

C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.

A₂₀: OK...OK. On Sunday I have ...

Properties of Human Conversation

- ▶ **Turn-taking issues**
 - ▶ When to take the floor?
 - ▶ When to yield the floor?
- ▶ **Interruptions**

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₄: And you're flying into what city?
C₅: Seattle.
A₆: And what time would you like to leave Pittsburgh?
C₇: Uh hmm I don't think there's many options for non-stop.
A₈: Right. There's three non-stops today.
C₉: What are they?
A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time.
The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the
last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₁₁: OK I'll take the 5ish flight on the night before on the 11th.
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C₁₃: OK.
A₁₄: And you said returning on May 15th?
C₁₅: Uh, yeah, at the end of the day.
A₁₆: OK. There's #two non-stops ... #
C₁₇: #Act... actually #, what day of the week is the 15th?
A₁₈: It's a Friday.
C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK...OK. On Sunday I have ...

Implications for Conversational Agents

- ▶ **Barge-in**

- ▶ Allowing the user to interrupt

- ▶ **End-pointing**

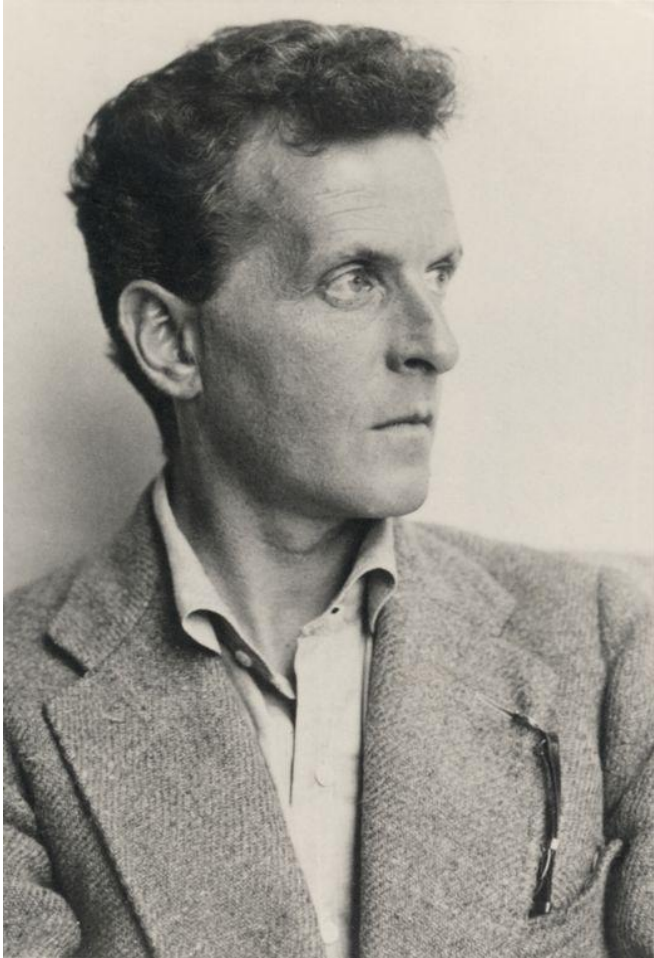
- ▶ The task for a speech system of deciding whether the user has stopped talking.
 - ▶ Very hard, since people often pause in the middle of turns

Implications for Conversational Agents

In a dialogue system, recognizing when the user is talking is called barge-in.

A system also has to know when to start talking. For example, most of the time in conversation, speakers start their turns almost immediately after the other speaker finishes, without a long pause. This is sometimes called "latching"; people are able to (most of the time) detect when the other person is about to finish talking and time their response to slip right in.

Language as Action



Each turn in a dialogue is a kind of action
Wittgenstein (1953) and Austin (1962)

A key insight into conversation—due originally to the philosopher Wittgenstein (1953) but worked out more fully by the philosopher Austin (1962)—is that each utterance in a dialogue is a kind of action being performed by the speaker.

Speech Acts (aka Dialogue Acts) Bach and Harnish 1979.

Constatives: committing the speaker to something's being the case (*answering, claiming, confirming, denying, disagreeing, stating*)

Directives: attempts by the speaker to get the addressee to do something (*advising, asking, forbidding, inviting, ordering, requesting*)

Commissives: committing the speaker to some future course of action (*promising, planning, vowing, betting, opposing*)

Acknowledgments: express the speaker's attitude regarding the hearer with respect to some social action (*apologizing, greeting, thanking, accepting an acknowledgment*)

Speech acts

- ▶ "Turn up the music!"

DIRECTIVE

- ▶ "What day in May do you want to travel?"

DIRECTIVE

- ▶ "I need to travel in May"

CONSTATIVE

- ▶ Thanks

ACKNOWLEDGEMENT

Grounding

- Participants in conversation or any joint activity need to establish **common ground**.
- **Principle of closure.** Agents performing an action require evidence, sufficient for current purposes, that they have succeeded in performing it (Clark 1996, after Norman 1988)
- Speech is an action too! So speakers need to **ground** each other's utterances.
 - **Grounding:** acknowledging that the hearer has understood

Grounding

- Grounding is relevant for human-machine interaction
 - Why do elevator buttons light up?



Grounding: Establishing Common Ground

▶A: And you said returning on May 15th?

C: Uh, yeah, at the end of the day.

A: **OK**

▶C: OK I'll take the 5ish flight on the night before on the 11th.

A: **On the 11th? OK.**

▶C: ...I need to travel in May.

A: **And**, what day **in May** did you want to travel?

Grounding is important for computers too!

▶System: Did you want to review some more of your profile?

▶User: No.

▶System: What's next?

Awkward!

▶System: Did you want to review some more of your profile?

▶User: No.

▶System: **Okay**, what's next?

Less Awkward!

Conversations have structure

- ▶ Local structure between adjacent speech acts, from the field of **conversational analysis** (Sacks et al. 1974)
- ▶ Called **adjacency pairs**:
- ▶ QUESTION... ANSWER
- ▶ PROPOSAL... ACCEPTANCE/REJECTION
- ▶ COMPLIMENTS ("Nice jacket!")... DOWNPLAYER ("Oh, this old thing?")

Another kind of structure: Subdialogues

▶ **Correction** subdialogue

▶ **Agent:** OK. There's #two non-stops#

Client: #Act- actually#, what day of the week is the 15th?

Agent: It's a Friday.

Client: Uh hmm. I would consider staying there an extra day til Sunday.

▶ **Agent:** OK...OK. On Sunday I have ...

Clarification Subdialogues

► **User:** What do you have going to UNKNOWN WORD on the 5th?

System: Let's see, going where on the 5th?

User: Going to Hong Kong.

► **System:** OK, here are some flights...

Presequences

- ▶ **User:** Can you make train reservations?
- ▶ **System:** Yes I can.
- ▶ **User:** Great, I'd like to reserve a seat on the 4pm train to New York.

Practice code

- ▶ 1. **Add a New Rule**: Add a new rule to the rules list to make the chatbot respond with "That's a great question!" when the user says "why".
- ▶ 2. **Modify a Response**: Change the `get_name()` method to make the chatbot respond with "My name is" followed by its name and "How can I assist you today?".

Reference

- ▶ <https://web.njit.edu/~ronkowitz/eliza.html>
- ▶ <https://anthay.github.io/eliza.html>
- ▶ <https://www.youtube.com/watch?v=zhxNI7V2IxM>

Question



Thank you

